AP Physics 1

Name _____

Period _____ Date _____

Practice Test – Introductory Dynamics (Version 5)

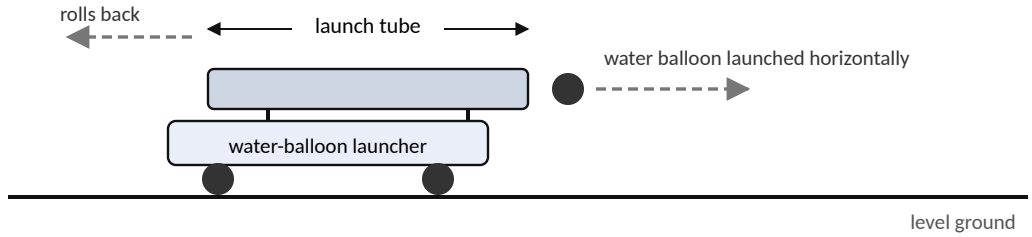
On any problems, you must show all work, include units, round properly, and box your final answer. Do the algebra first, before plugging in values.

Rewrite the following statement in the space below: On my honor, before God, I have neither given, nor received, any aid on this test.

For all questions:

- Use g = 9.8 m/s² (9.8 N/kg). Air resistance is negligible unless the question says it is not.
- Free-body diagrams: draw the forces themselves, not components, each as an arrow starting on the dot.
- Round answers to the significant figures of the given values.
- 50 minutes. Four questions, 25 points each (100 points total).

1. (25 points) A water-balloon launcher on a golf cart launches a water balloon horizontally to the right. Find the water-balloon launcher's acceleration while the water balloon is in the tube.



water balloon mass	450. g
mass of the launcher and golf cart	450. kg
launch tube length	120. cm
exit speed, relative to the ground	54.0 km/h

The golf cart moves back less than 2 mm while the balloon is in the tube, so use the tube length as the distance the balloon moves relative to the ground. This is an approximation.

Assumptions for this practice problem: *the water-balloon launcher starts at rest on level ground and rolls freely (ignore friction and air resistance); the water balloon starts from rest; its acceleration in the tube is constant; the push between the water balloon and the water-balloon launcher is the only horizontal interaction.*

(a) Draw the free-body diagram of the water-balloon launcher while the water balloon is in the tube.

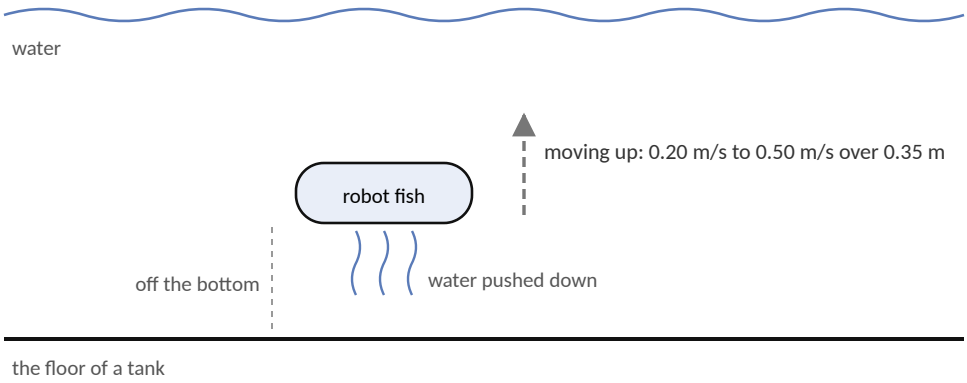
(a) water-balloon launcher

(b) Calculate the water balloon's acceleration relative to the ground while the water balloon is in the tube.

(c) Calculate the magnitude of the force the water-balloon launcher exerts on the water balloon, and the magnitude of the force the water balloon exerts on the water-balloon launcher.

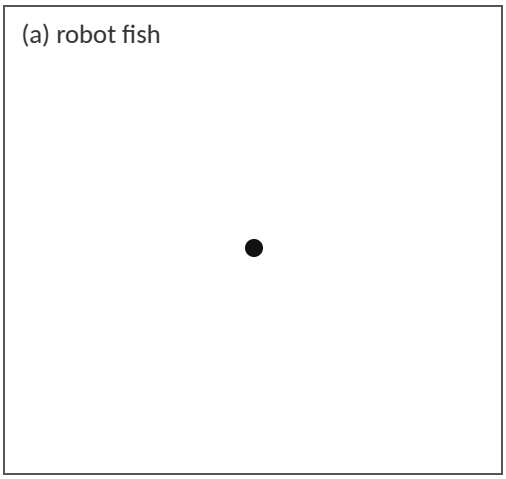
(d) Calculate the water-balloon launcher's acceleration, magnitude and direction.

2. (25 points) A 0.360 kg robot fish has lifted off the floor of a tank. Its fins push water downward, so the water pushes it upward with a thrust of 0.60 N. After it leaves the bottom, moving upward, it speeds up from 0.20 m/s to 0.50 m/s while rising 0.35 m.



Assumptions for this practice problem: *water drag on the robot fish is negligible; its acceleration is constant; g = 9.8 m/s².*

(a) Draw the free-body diagram of the robot fish after it has left the bottom.

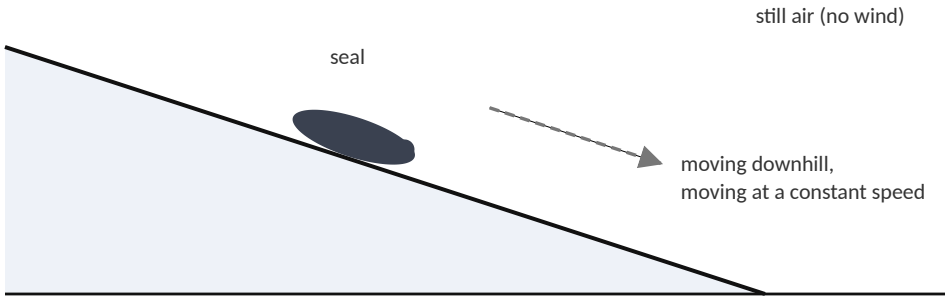


(b) Calculate the robot fish's acceleration, magnitude and direction.

(c) With up as positive, write Newton's second law for the robot fish and solve it for the buoyant force in letters.

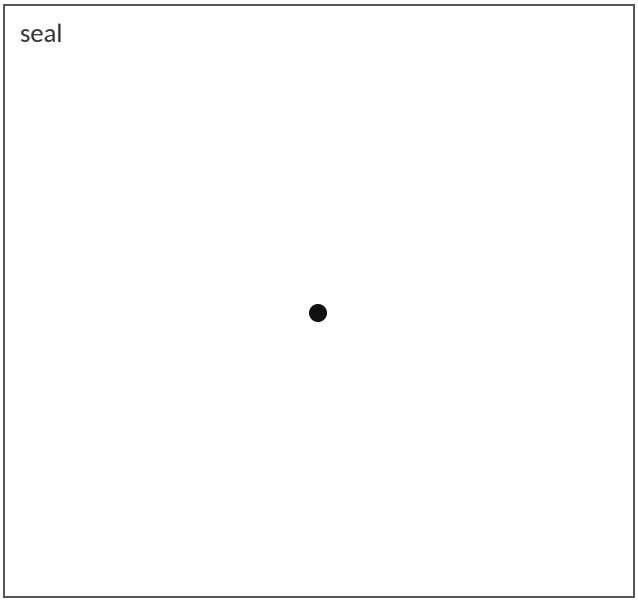
(d) Calculate the magnitude of the buoyant force on the robot fish.

3. (25 points) A seal slides on its belly down an icy bank. In the picture the seal moves down the slope to the right and is moving at a constant speed. Friction from the ice and drag from the air are **not** negligible. No angle is given.



Assumptions for this practice problem: *the air is still (no wind), so the air moves past the seal opposite its motion.*

Draw and label the free-body diagram of the seal. Draw the forces themselves, not components. Each force is one arrow that starts on the dot and points in the direction the force acts. No calculations.



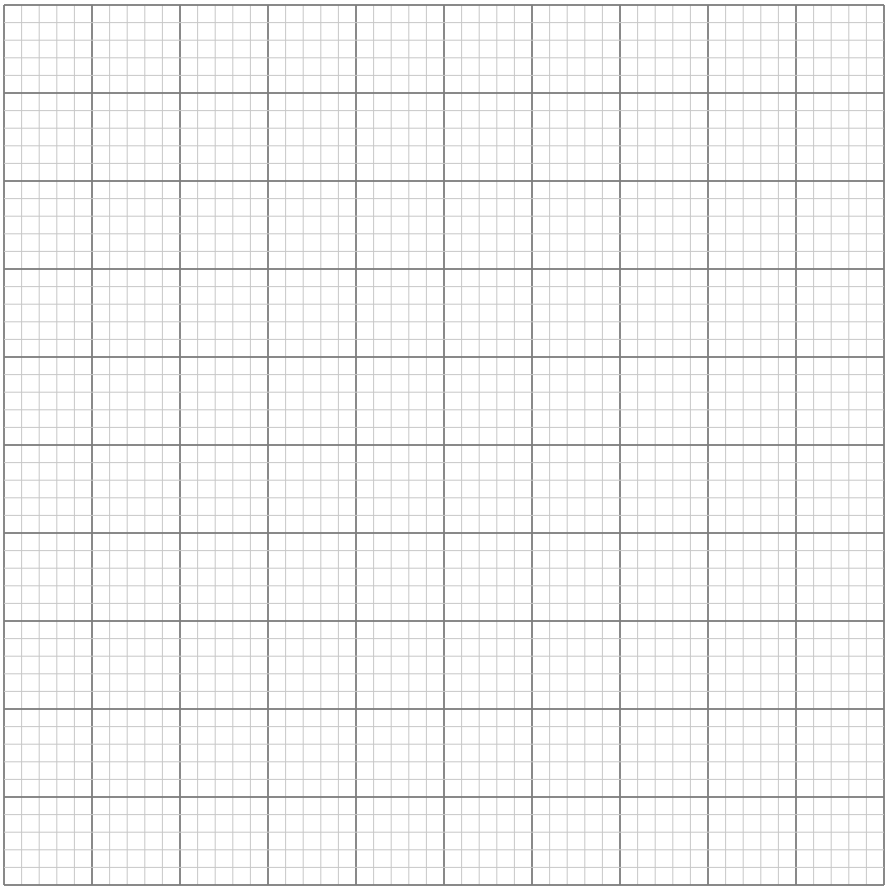
4. (25 points) Students measure the drag force F_D on a drone body in a wind tunnel at several air speeds v. The drag force is modeled by $F_D = Kv^2$, where K is a constant.

Speed v (m/s)	**Drag force F_D (N)**	
5.00	0.534	
9.00	1.71	
13.0	3.61	
17.0	6.04	
21.0	8.92	
25.0	12.8	

(a) Using $F_D = Kv^2$, state the quantity you will plot on the vertical axis and the quantity you will plot on the horizontal axis so the data form a straight line, each with its units. State what the slope equals.

(b) Use the blank column in the table to calculate the values you will plot on the horizontal axis.

(c) Plot the data on the grid. Label each axis with the quantity and units, choose a scale, and draw the best-fit line.



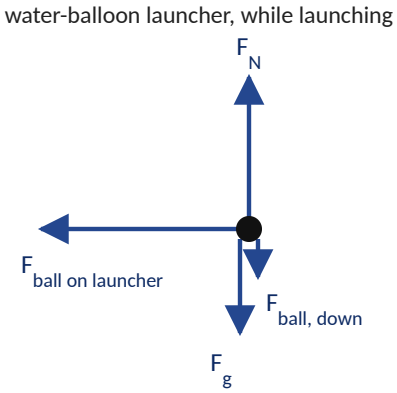
(d) Calculate the slope of your best-fit line using two well-separated points on the line. Then give K with its units.

Answer Key – Practice Test, Version 5 – Introductory Dynamics

Worked step by step: diagram, symbolic equations and algebra first, then numbers with units. $g = 9.8\ m/s^2$ does not limit significant figures; answers match the significant figures of the givens.

Numerical problems: diagram first, then a = ΣF/m (equation sheet), written $a = F_{net}/m$. Replace F_{net} with the actual forces and their signs, swap $F_g = mg$, solve in letters, then plug in numbers with units.
Kinematics: #1 $v = v_0 + at$ · #2 $\Delta x = v_0 t + \frac{1}{2}at^2$ · #3 $v^2 = v_0^2 + 2a\Delta x$. Pick the one that leaves out the variable you don't know and don't need.

1. Recoil (25 points)



Step 1 **(a) Forces on the water-balloon launcher while the water balloon is in the tube.**

Vertical: the ground pushes up (F_N); Earth pulls down (F_g); the water balloon presses down ($F_{ball,\ down}$). Nothing accelerates vertically, so $F_N = F_g + F_{ball,\ down}$.

Horizontal: the only force is the water balloon pushing back on the water-balloon launcher, $F_{ball\ on\ launcher}$, to the left. The vertical forces do not enter the horizontal net-force equation. Arrow lengths are not to scale.

Step 2 **Units and frames.**

Convert first: 450. g = 0.450 kg; 120. cm = 1.20 m;

$$54.0\ km/h \times \frac{1000\ m}{1\ km} \times \frac{1\ h}{3600\ s} = 15.0\ m/s.$$

Step 3 **(b) Kinematics for the water balloon.**

Balloon, relative to the ground: $v_0 = 0$, $v = 15.0$ m/s, $\Delta x \approx 1.20$ m (tube length, as stated). Time is not given: use #3.

$$v^2 = v_0^2 + 2a\Delta x \rightarrow a_{ball} = \frac{v^2}{2\Delta x} = \frac{(15.0\ m/s)^2}{2(1.20\ m)} = \frac{225\ m^2/s^2}{2.40\ m} = 93.75\ m/s^2 \rightarrow \boxed{93.8\ m/s^2}$$

Step 4 **(c) Second law on the water balloon, then the third law.**

$$a = \frac{F_{net}}{m} \rightarrow F_{launcher\ on\ ball} = m_{ball}a_{ball} = (0.450\ kg)(93.75\ m/s^2) = 42.19\ N \rightarrow$$

42.2 N, forward on the water balloon

Third law: $F_{ball\ on\ launcher} = -F_{launcher\ on\ ball} \rightarrow$ **42.2 N, backward on the water-balloon launcher**

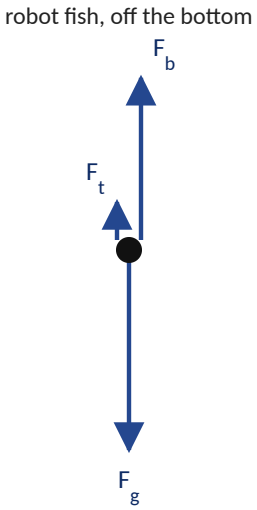
Step 5 **(d) Second law on the water-balloon launcher, horizontal only.**

$$a_{launcher} = \frac{F_{ball\ on\ launcher}}{M} = \frac{42.19\ N}{450.\ kg} = \boxed{0.0938\ m/s^2,\ \text{to the left (backward)}}$$

Check the approximation: $t = v/a_{ball} = 0.160$ s, so the golf cart moves $\frac{1}{2}(0.0938\ m/s^2)(0.160\ s)^2 = 0.0012$ m, about 0.1% of the tube.

Watch out: convert first: 450. g = 0.450 kg, 120. cm = 1.20 m, 54.0 km/h = 15.0 m/s.

2. Thrust and buoyancy (25 points)



Step 1 **(a) Forces on the robot fish after it leaves the bottom.**

F_g down (Earth). Thrust F_t up (the water it pushes down pushes back up on it). Buoyancy F_b up (the surrounding water).

No normal force: it is no longer touching the bottom. Acceleration is up, so $F_t + F_b$ is longer than F_g.

Step 2 **(b) Acceleration from the motion data.**

Two speeds and a displacement, no time: use #3.

$$v^2 = v_0^2 + 2a\Delta y \rightarrow a = \frac{v^2 - v_0^2}{2\Delta y} = \frac{(0.50\ \text{m/s})^2 - (0.20\ \text{m/s})^2}{2(0.35\ \text{m})} = \boxed{0.30\ \text{m/s}^2\text{, upward}}$$

Step 3 **(c) Newton's second law, up is +, letters first.**

$$a = \frac{F_{net}}{m} \rightarrow a = \frac{F_b + F_t - F_g}{m} \rightarrow F_b + F_t - mg = ma$$

$$\boxed{F_b = m(g + a) - F_t}$$

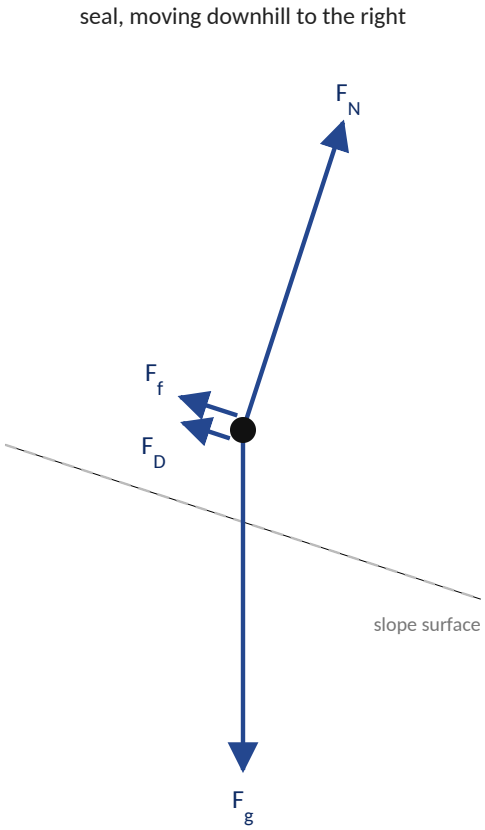
Step 4 **(d) Now the numbers, units in every parenthesis.**

$F_b = (0.360\ \text{kg})(9.8\ \text{m/s}^2 + (0.30\ \text{m/s}^2)) - 0.60\ \text{N} = (0.360\ \text{kg})(10.1\ \text{m/s}^2) - 0.60\ \text{N} = 3.636\ \text{N} - 0.60\ \text{N} = 3.036\ \text{N}$

$\boxed{F_b = 3.04\ \text{N}}$ (the subtraction sets the decimal place; g does not limit the significant figures).

Watch out: keep the sign of a. Using a = 0 just because the robot fish moves up, or dropping the thrust, gives the wrong buoyant force.

3. Free-body diagram on a slope (25 points)



Step 1 **List every force on the seal: exactly four.**

Earth pulls on it: F_g. The ice pushes on it two ways: F_N (perpendicular to the surface) and kinetic friction F_f (along the surface). The air pushes on it: drag F_D.

Step 2 **Directions. No angle is needed.**

F_g: straight down, vertical, not perpendicular to the slope.

F_N: perpendicular to the slope, pointing out of the surface, so it tilts away from vertical toward the right.

F_f: along the slope, uphill (up and to the left), opposite the sliding.

F_D: opposite the motion through the air. The air is still and the seal moves downhill, so drag also points uphill, parallel to F_f.

Step 3 **Check the diagram.**

Four arrows, each starting on the dot, each labeled. F_f and F_D are two separate arrows, side by side, both uphill. Speeding up, slowing down or moving at a constant speed changes the arrow lengths, not their directions.

Lengths in this key: constant speed, so the two uphill arrows together balance the part of gravity that pulls along the slope, and FN is a little shorter than Fg. Lengths are not graded.

Watch out: do not add the components of F_g (along and perpendicular to the slope) as extra arrows; they are not new forces. Drag is the air resistance, not a fifth force, and there is no forward "motion" force.

4. Drag linearization (25 points)

Step 1 **(a) Match $F_D = Kv^2$ to $y = mx + b$.**

$F_D = K \cdot (v^2) + 0$

vertical (y)	horizontal (x)	slope	intercept
F_D (N)	v^2 (m^2/s^2)	K	0

Vertical: F_D in N. Horizontal: v^2 in m^2/s^2. Slope = K.

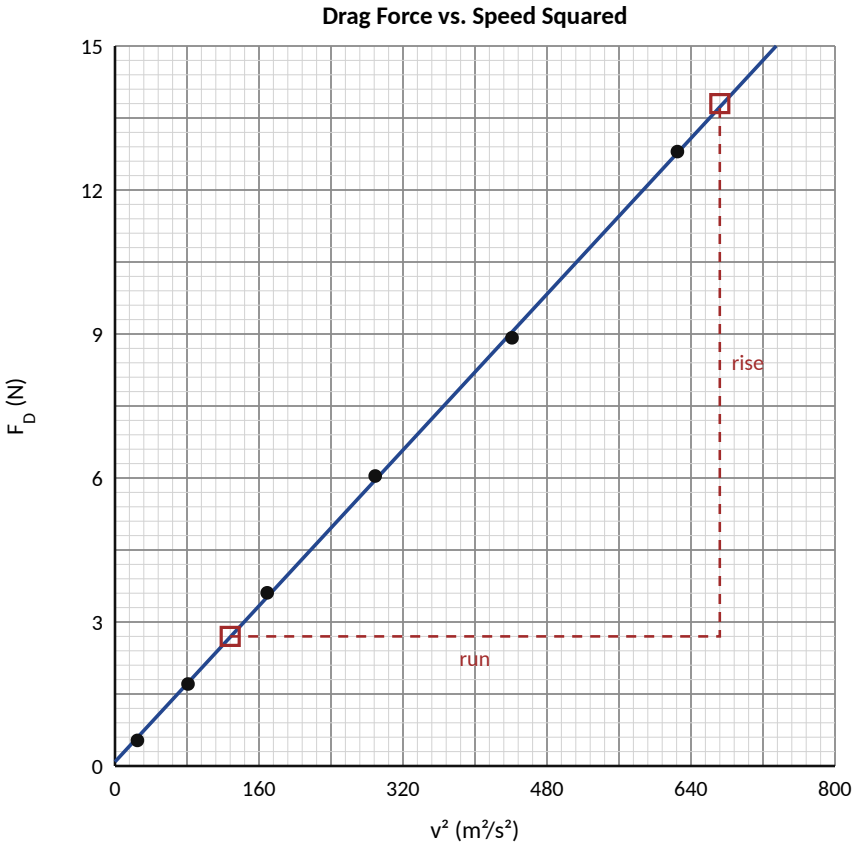
Watch out: v^2 is speed squared, in m^2/s^2. It is not an acceleration (m/s^2).

Step 2 **(b) Square each speed; the units square too.**

Example: $(5.00\ m/s)^2 = 25.0\ m^2/s^2$

v (m/s)	F_D (N)	v^2 (m^2/s^2)
5.00	0.534	25.0
9.00	1.71	81.0
13.0	3.61	169
17.0	6.04	289
21.0	8.92	441
25.0	12.8	625

Step 3 **(c) Graph: labeled axes with units, a scale that uses more than half the grid, one straight line through the middle of the points.**



The best-fit line is not forced through any data point. The red squares are points read off the line, not data points.

Step 4 **(d) Slope from two well-separated points on the line, units carried.**

Points read off the line, away from data points: (128 m^2/s^2, 2.70 N) and (672 m^2/s^2, 13.80 N)

$$\text{slope} = \frac{13.80\ \text{N} - 2.70\ \text{N}}{672\ \text{m}^2/\text{s}^2 - 128\ \text{m}^2/\text{s}^2} = \frac{11.10\ \text{N}}{544\ \text{m}^2/\text{s}^2} = 0.020404\ \text{N}\cdot\text{s}^2/\text{m}^2$$

$K \approx 0.0204\ \text{N}\cdot\text{s}^2/\text{m}^2\ (= \text{kg/m})$. Any value from 0.0187 to 0.0219 $N\cdot s^2/m^2$ from a reasonable line earns credit.

Units: $N \div (m^2/s^2) = N\cdot s^2/m^2$, and $N = kg\cdot m/s^2$, so $N\cdot s^2/m^2 = kg/m$.

1. Recoil

Suggested partial credit	pts
(a) F_N up, F_g down (the water balloon pressing down may also be shown); backward push from the water balloon; no forward force	5
Unit conversions (or none needed) and the matching ground-frame values	3
(b) Kinematics equation chosen and solved in letters; $a_{ball} = 93.8\ m/s^2$	5
(c) $F = m_{ball}a_{ball} = 42.2$ N, and the third law gives the same size back on the water-balloon launcher	5
(d) acceleration = F/M = 0.0938 m/s^2, backward	7
Total	**25**

2. Thrust and buoyancy

Suggested partial credit	pts
(a) F_g down, thrust and buoyancy up, labeled	4
(a) No normal force	2
Acceleration: correct kinematics equation, magnitude and direction (sign)	6
$F_b + F_t - mg = ma$ written with up as +	4
Solved in letters: $F_b = m(g + a) - F_t$	3
Substitution with units in every parenthesis (and grams to kg if needed)	3
$F_b = 3.04$ N	3
Total	**25**

3. Free-body diagram on a slope

Suggested partial credit	pts
F_g straight down (vertical), labeled	5
F_N perpendicular to the slope, out of the surface, labeled	5
F_f along the slope, uphill, labeled	5
F_D uphill (opposite the motion through still air), labeled, separate from F_f	5
Every arrow starts on the dot; no components and no extra forces	5
Total	**25**

4. Drag linearization

Suggested partial credit	pts
(a) F_D (N) on the vertical axis, v^2 (m^2/s^2) on the horizontal axis	4
(a) Slope = K	2
(b) v^2 column correct, with units	4
(c) Axes labeled with quantity and units	2
(c) Scale uses more than half the grid	2
(c) Points plotted correctly	2
(c) One straight best-fit line	2
(d) Two well-separated points read off the line, not data points	3
(d) K from 0.0187 to 0.0219, with units $N \cdot s^2/m^2$ (or kg/m)	4
Total	**25**